INTRODUCTION TO SQL SERVER

Objectives

- SQL, T-SQL
- Hardware and Software Requirements
- MS SQL Server 2012 Edition
- MS SQL Server 2012 Installation
- MS SQL Server 2012 Tools and Components
- System Database

Structured Query Language (SQL)

- **Is an industry-standard programming language used to insert, retrieve, modify, and delete data in a relational database.**

- **contains statements for defining and administering the objects in a database**

Structured Query Language (SQL)

- **is the language supported by almost all database systems, and is the subject of published industry standards**

- **developed by E. F. Codd while he was working at the IBM research labs in San Jose in 1971.**

Structured Query Language (SQL)

Three Categories:

1. Data Definition Language (DDL)

- used to define and manage all the objects in an SQL Server such as databases, tables and view.

- CREATE, ALTER and DROP statements

Three Categories:

2. Data Manipulation Language (DML)

- **used to SELECT, INSERT, UPDATE and DELETE data in the objects defined by DDL**

Three Categories:

3. Data Control Language (DCL)

- **Security commands such as GRANT, REVOKE, and DENY that control how a principal (user or role) can access a securable (object or data.)**

Transact-SQL (T-SQL)

- T-SQL is the language containing the commands used to administer SQL Server, to create and manage all objects in SQL Server, and to insert, retrieve, modify, and delete all data in tables.

Microsoft SQL Server a History

Version	Year	Release Name	Codename
4.21 (WinNT)	1993	SQL Server 4.21	SQLNT
6.0	1995	SQL Server 6.0	SQL95
6.5	1996	SQL Server 6.5	Hydra
7.0	1999	SQL Server 7.0	Sphinx
-	1999	SQL Server 7.0 OLAP Tools	Plato
8.0	2000	SQL Server 2000	Shiloh
8.0	2003	SQL Server 2000 64-bit Edition	Liberty
9.0	2005	SQL Server 2005	Yukon
10.0	2008	SQL Server 2008	Katmai
10.25	2010	SQL Azure DB	CloudDatabase
10.5	2010	SQL Server 2008 R2	Kilimanjaro
11.0	2012	SQL Server 2012	Denali
12.0	2014	SQL Server 2014	SQL14

SQL Server Architecture

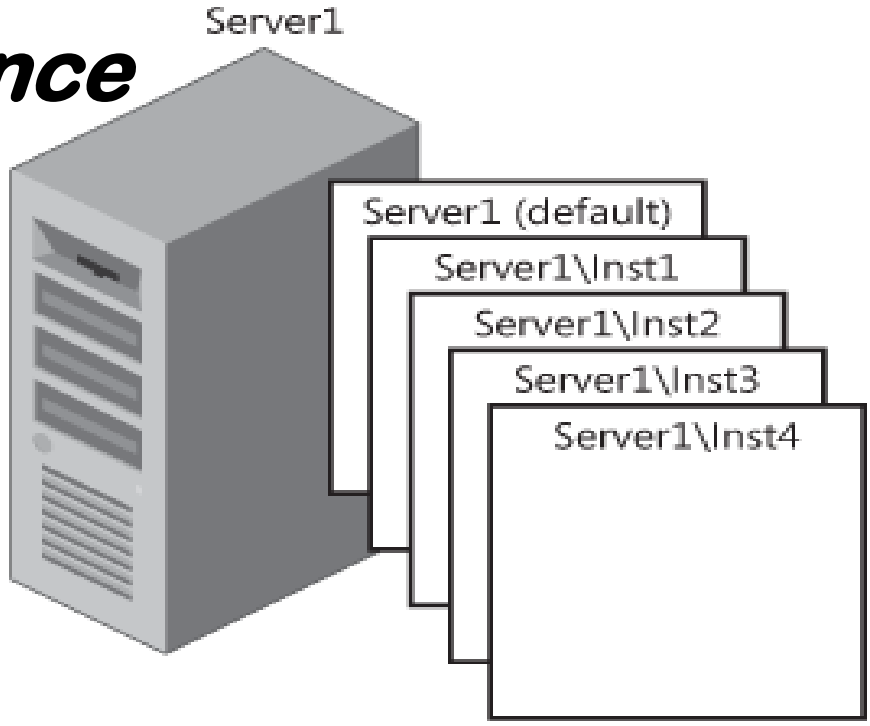
The entities in involved SQL Server architecture

- SQL Server instances,
- databases
- schemas
- database

SQL Server Architecture

SQL Server Instances

- is an installation of a SQL Server database engine/service databases
- can be a *Default Instance* or *Named Instance*



SQL Server Architecture

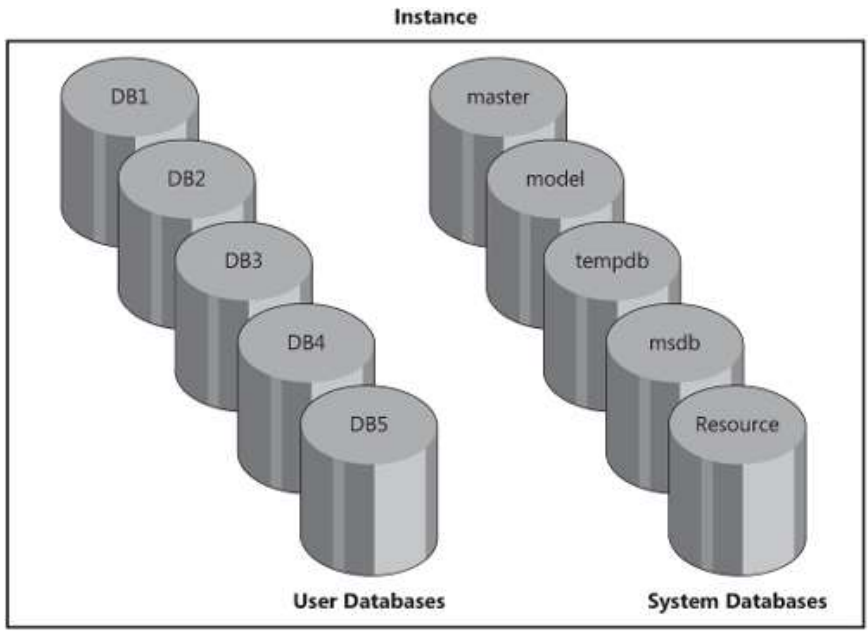
SQL Server Instances

- can install multiple instances of SQL Server on the same computer
- each instance is completely independent of the others in terms of security, the data that it manages, and in all other respects

SQL Server Architecture

Databases

- You can think of a database as a container of objects such as tables, views, stored procedures, and so on
- Each instance of SQL Server can contain multiple databases



MS SQL Server Edition

Enterprise Edition

- High-end edition with the advanced performance and availability features required to support thousands of connections and databases measured by terabytes

- This edition includes all the right features, including Integration Services, Analysis Services, Web Services, database mirroring, and failover clustering.

MS SQL Server Edition

Standard Edition

- perfectly suited for medium-size businesses that need all the primary features and can live without the very high-end high availability features of Enterprise Edition
- This edition includes all the right features, including Integration Services, Analysis Services, Web Services, database mirroring, and failover clustering.

MS SQL Server Edition

Workgroup Edition

- Intended as a departmental database server, Workgroup Edition includes the right mix of features for a small transactional database

Web Edition

- new for SQL Server 2012 is licensed for hosting websites

MS SQL Server Edition

Compact Edition

- Its small footprint of only 1MBof RAM means that it can actually run well on a mobile smart device

Express Edition

- It's a full version of the SQL Server database engine intended to serve as an embedded database within an application. It's completely free.

Hardware Requirements

Hardware	Requirements
Processor Speed	**Minimum:** x86 Processor: 1.0 GHz x64 Processor: 1.4 GHz **Recommended:** 2.0 GHz or faster
Memory	**Minimum:** Express Editions: 512 MB All other editions: 1 GB **Recommended:** Express Editions: 1 GB All other editions: At least 4 GB and should be increased as database size increases to ensure optimal performance.

Software Requirement

.NET 3.5 SP1 is a requirement for SQL Server 2012 when you select Database Engine, Reporting Services, Master Data Services, Data Quality Services, Replication, or SQL Server Management Studio, and it is no longer installed by SQL Server Setup.

.NET 4.0 is a requirement for SQL Server 2012. SQL Server installs .NET 4.0 during the feature installation step.

Windows PowerShell 2.0

OS Requirements

- Windows 7,
- Windows Server 2008 R2,
- Windows Server 2008 Service Pack 2
- Windows Vista Service Pack 2

Note: SQL Server 2012 installation on Windows 7 required to install Windows 7 SP1. It is recommended to perform Windows update before beginning SQL Server 2012 installation

Preparing to Install

1. Choosing Instance Name

- *Default Instance* - selected by default for the first installation of SQL Server on a machine

- *Named Instance* – used when a default instance already installed

Preparing to Install

2. Choosing Service Account



- **Local System**. The name of this account is NT AUTHORITY\System. It is a powerful account that has unrestricted access to all local system resources. It is a member of the Windows Administrators group on the local computer, and is therefore a member of the SQL Server sysadmin fixed server role
- **Network Service**. The name of this account is NT AUTHORITY\NetworkService. It is available in Microsoft Windows XP and Microsoft Windows Server 2003. All services that run under the Network Service account are authenticated to network resources as the local computer.

Preparing to Install

3. Selecting an Authentication Mode

Authentication Mode - is how SQL Server processes usernames and passwords

- *Windows Authentication Mode* - With this mode, a user can sit down at their computer, log in to the Windows domain, and gain access to SQL Server

Preparing to Install

3. Selecting an Authentication Mode

- *Mixed Mode (Windows Authentication Mode and SQL Server Authentication)*

  - The user logs in to their network, Windows or otherwise

  - The user opens a connection to SQL Server using a username and password other than those used to gain network access.

Authentication Modes

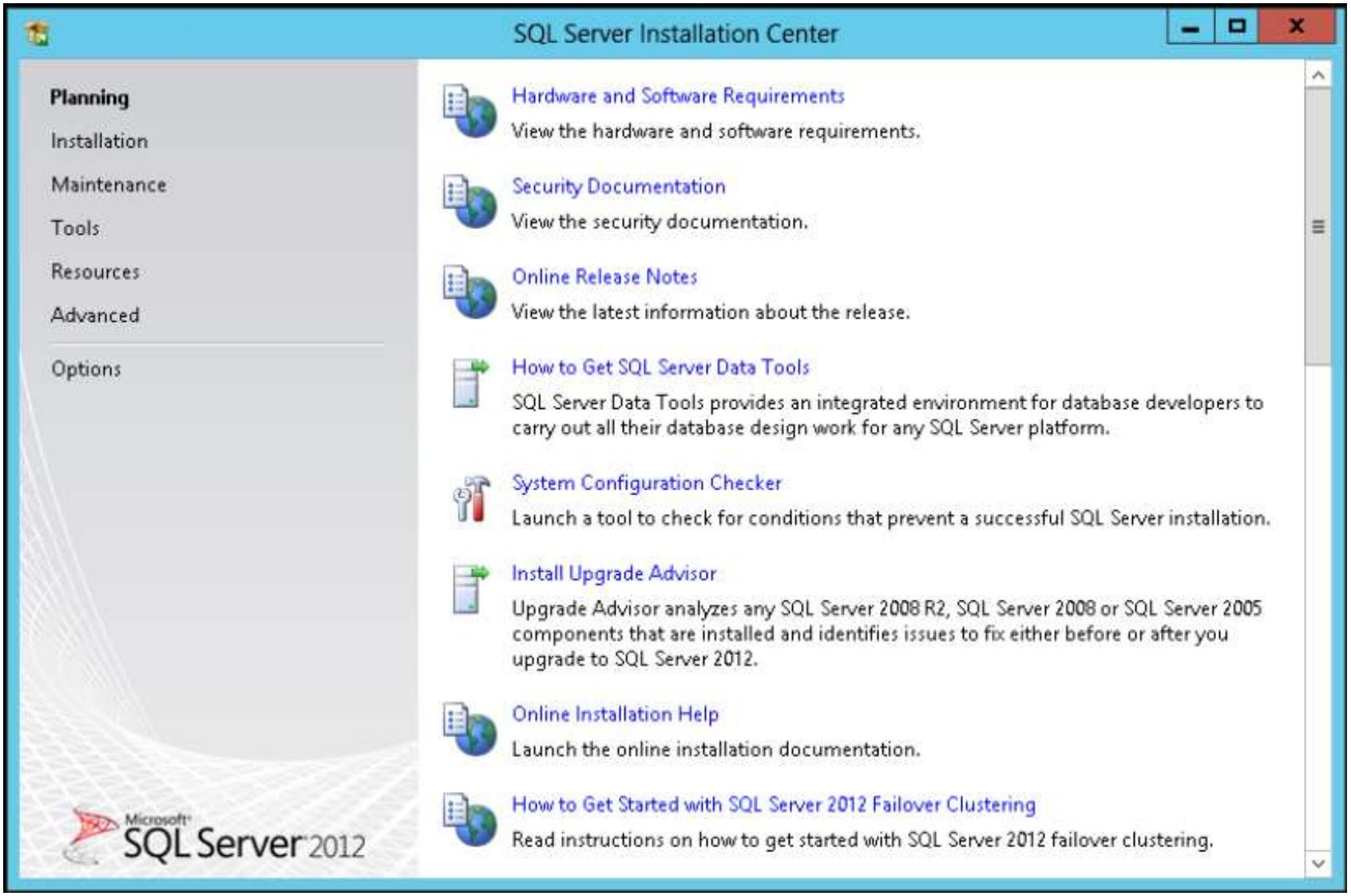
Security-Authentication Modes		
Login Method	*Windows Authentication Mode*	*Mixed Mode*
Users may authenticate using their Windows User logon	Yes	Yes
SQL Server–specific accounts	No	Yes

SQL Server 2012 Installation

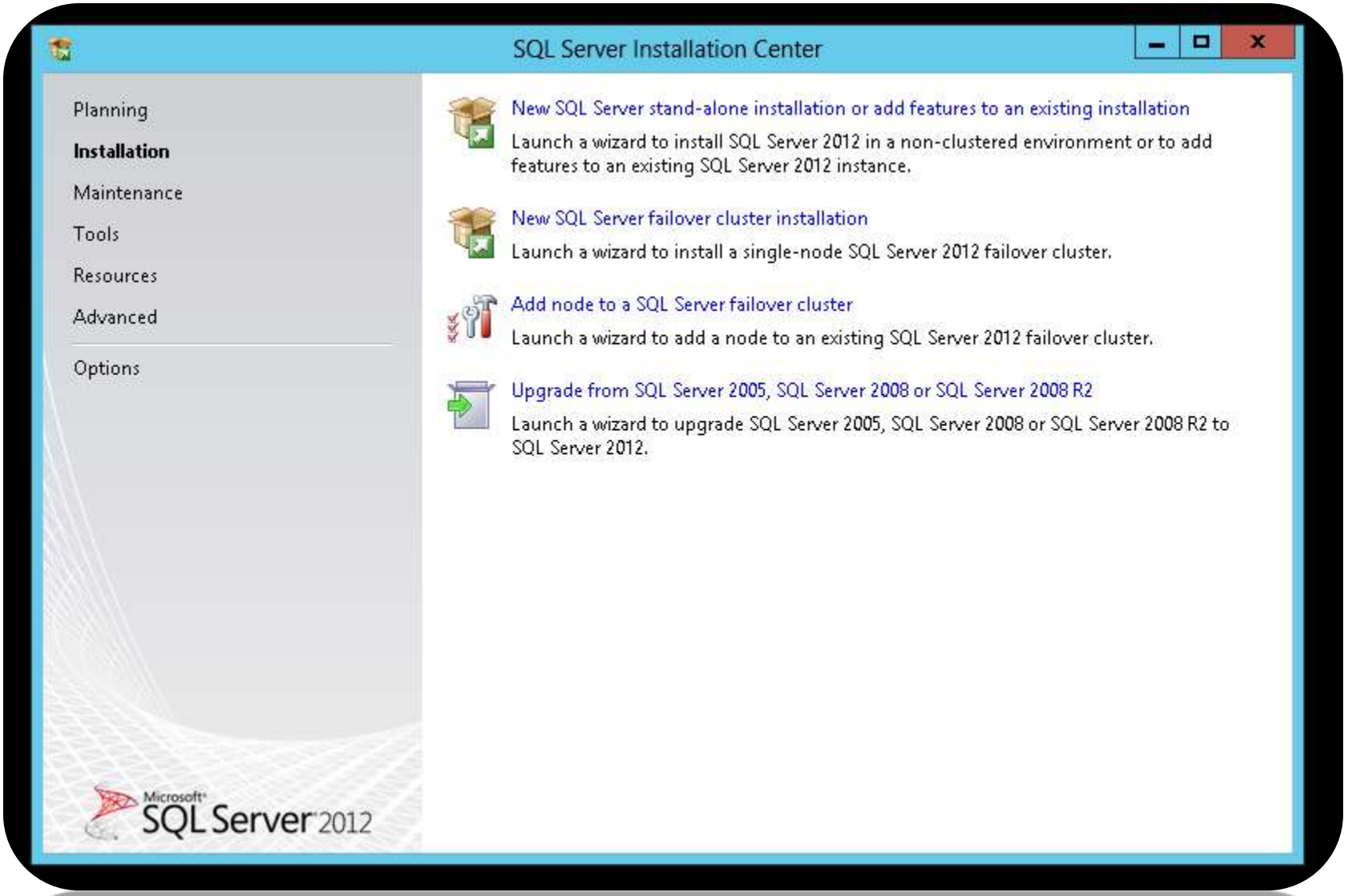
Installation – SQL Server 2012



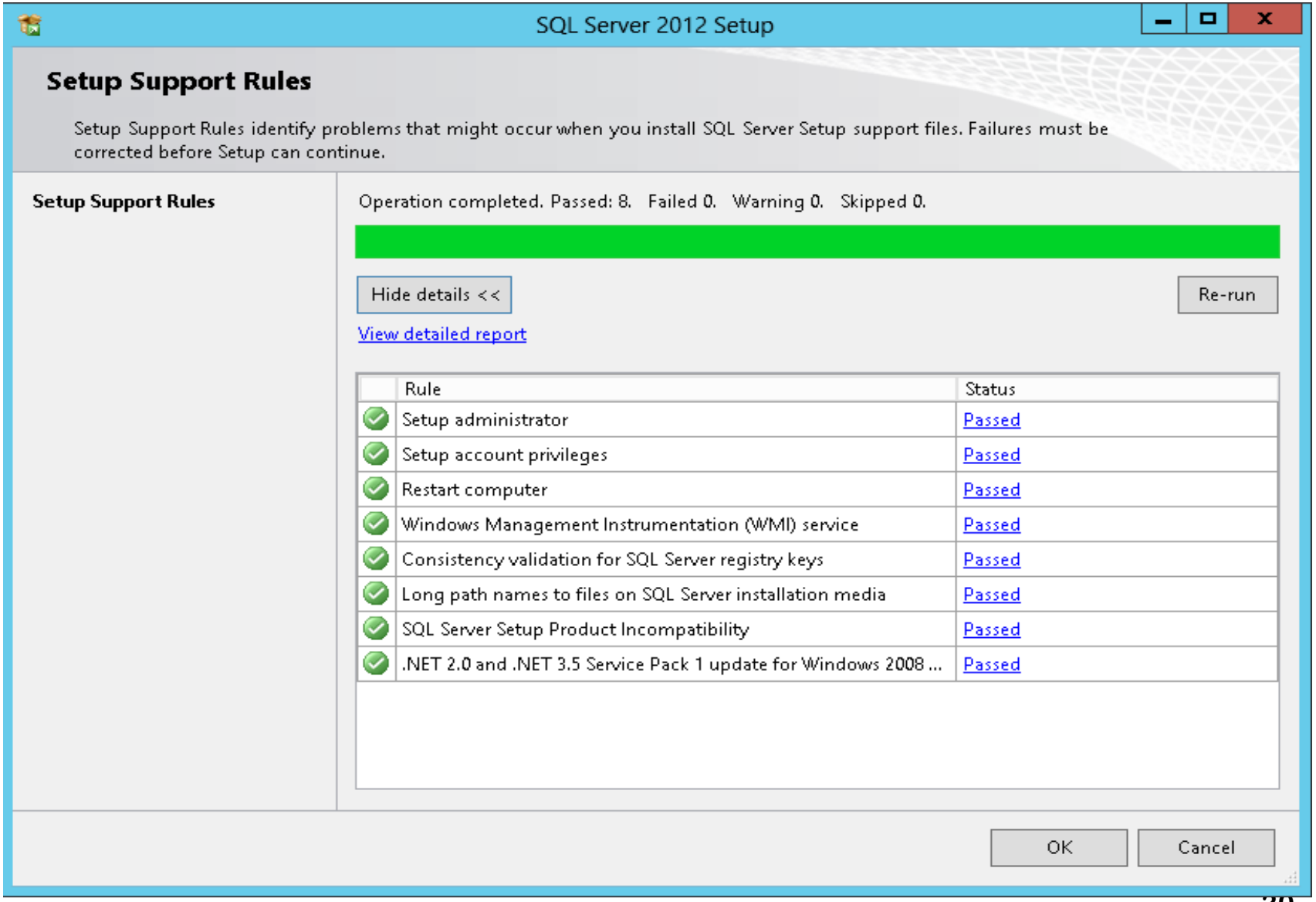
Installation – SQL Server 2012



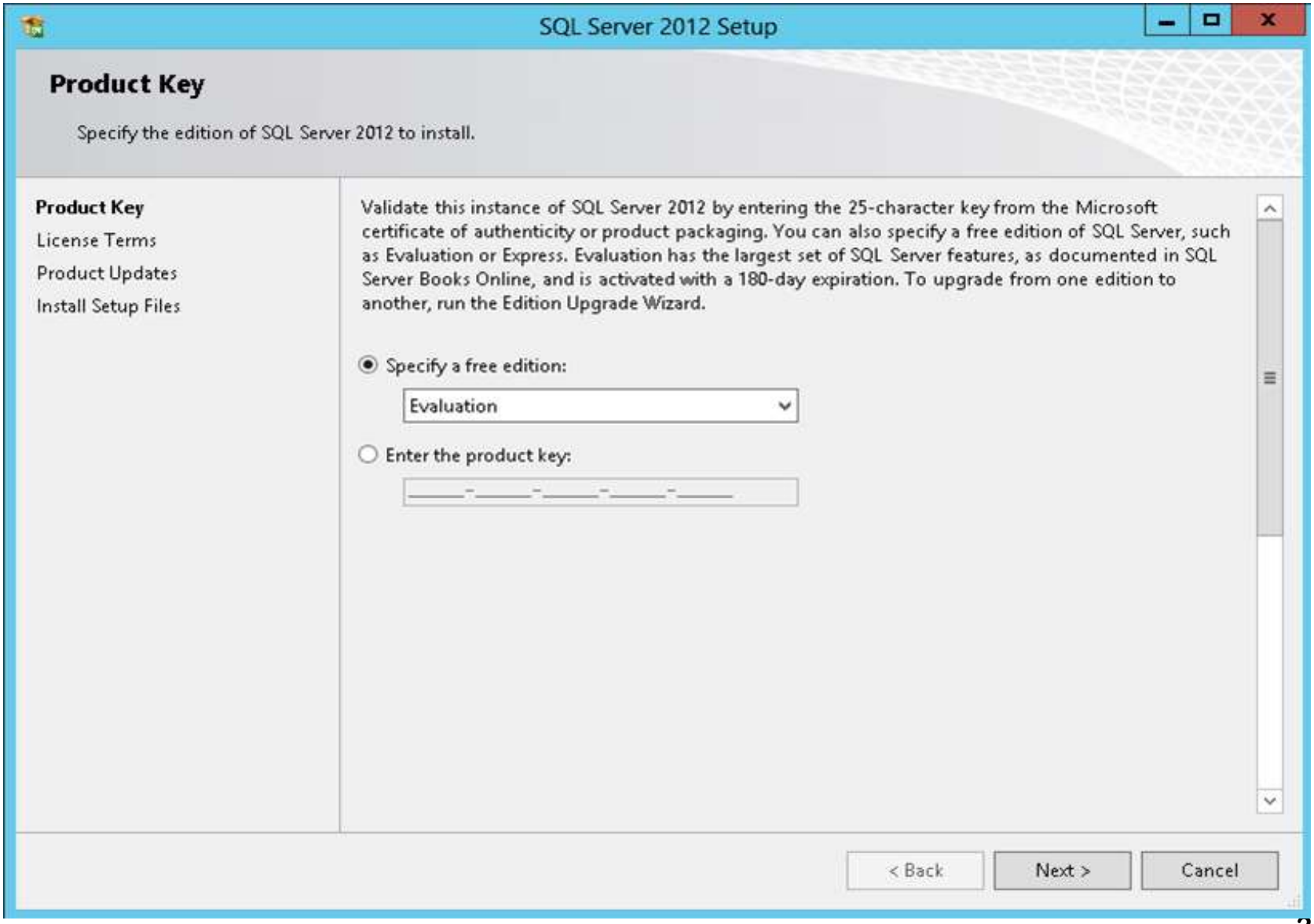
Installation – SQL Server 2012



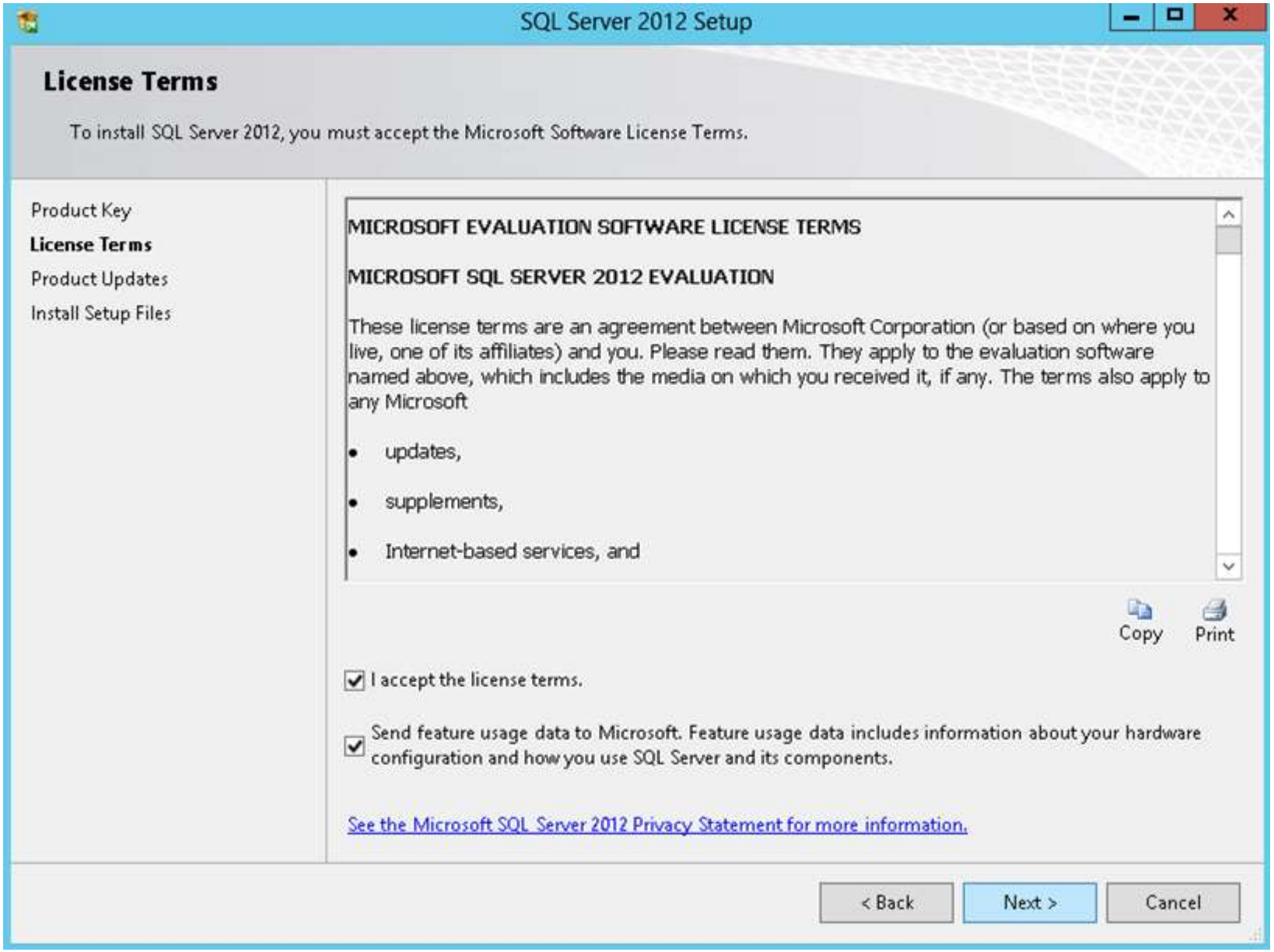
Installation – SQL Server 2012



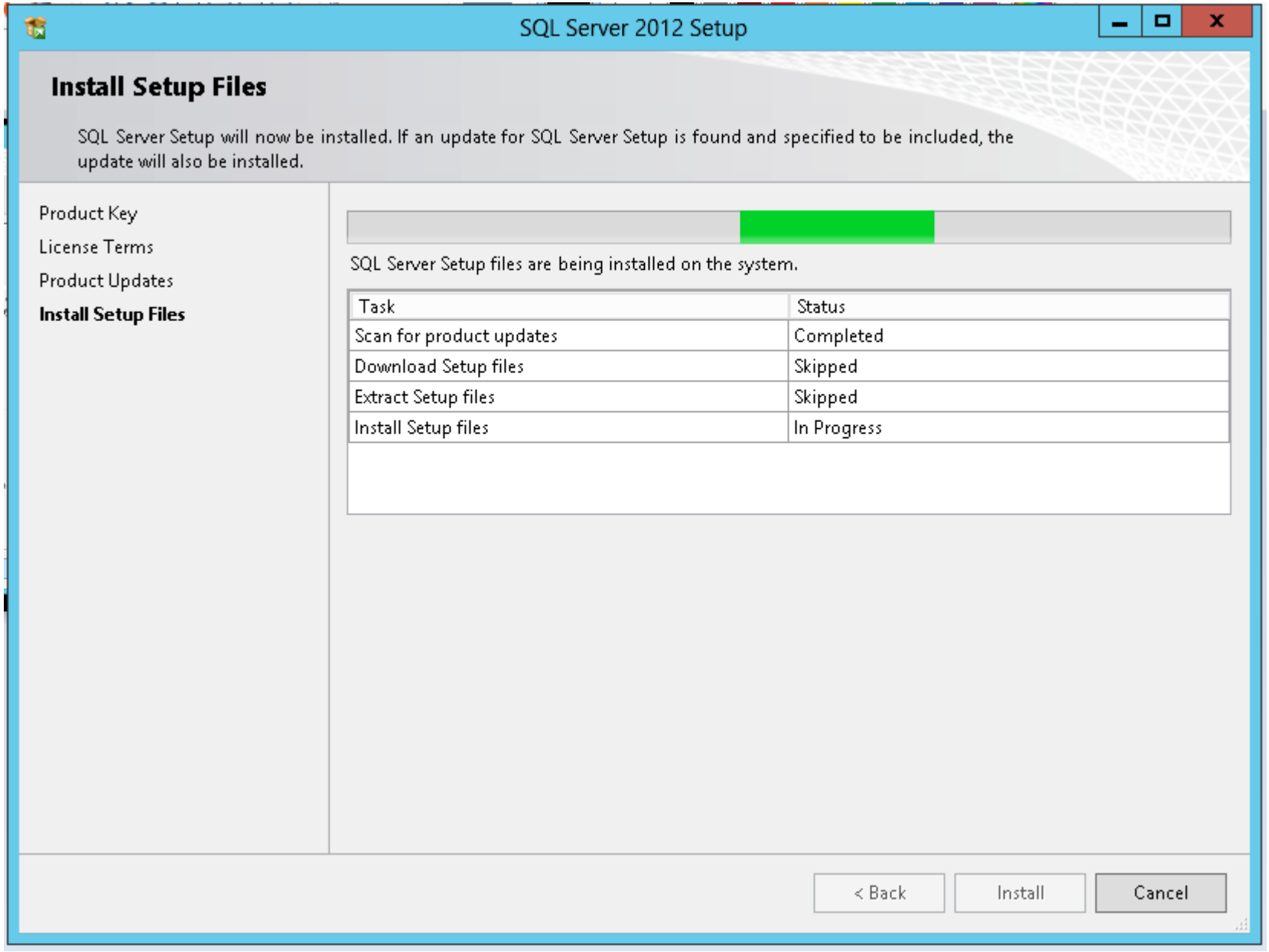
Installation – SQL Server 2012



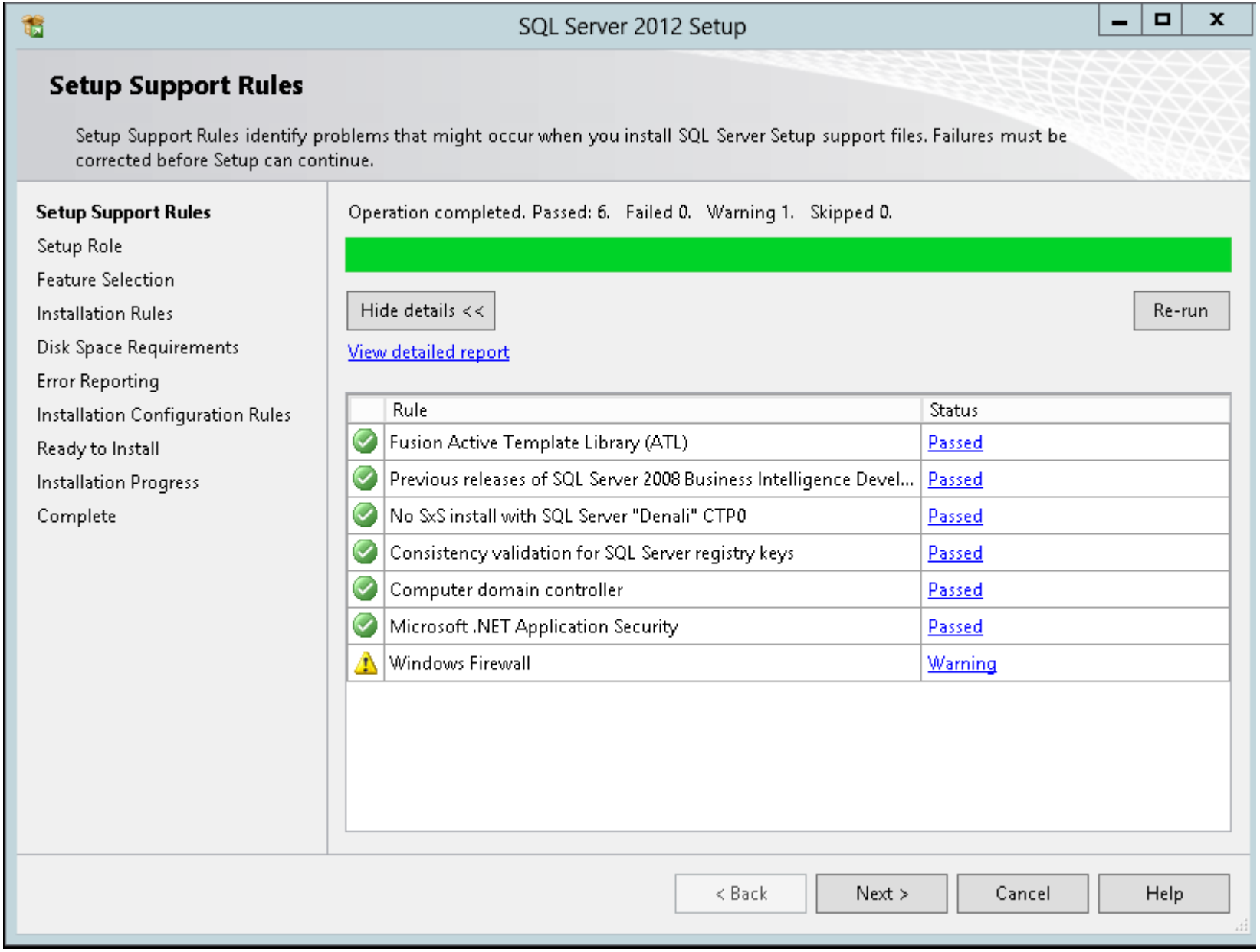
Installation – SQL Server 2012



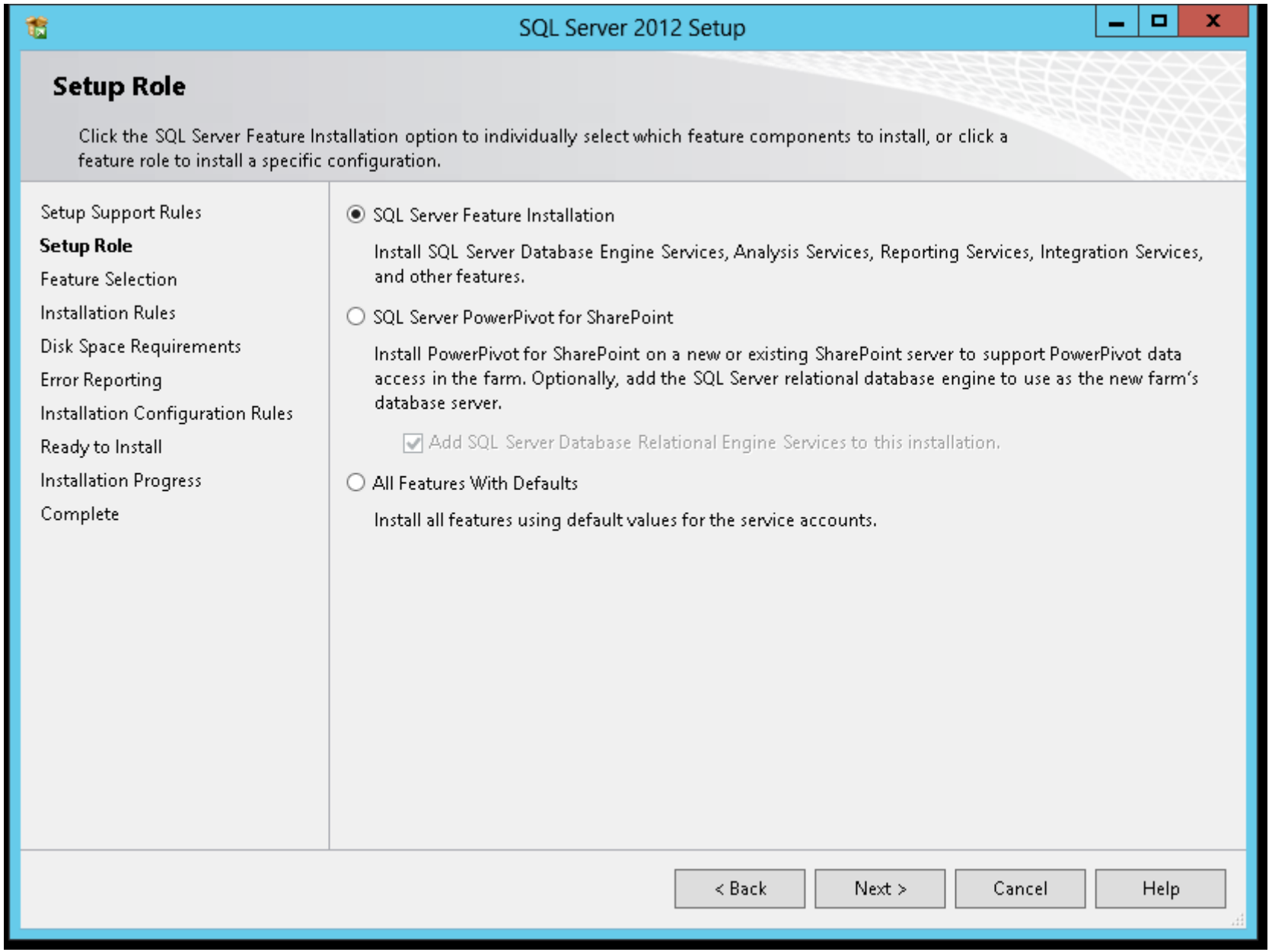
Installation – SQL Server 2012



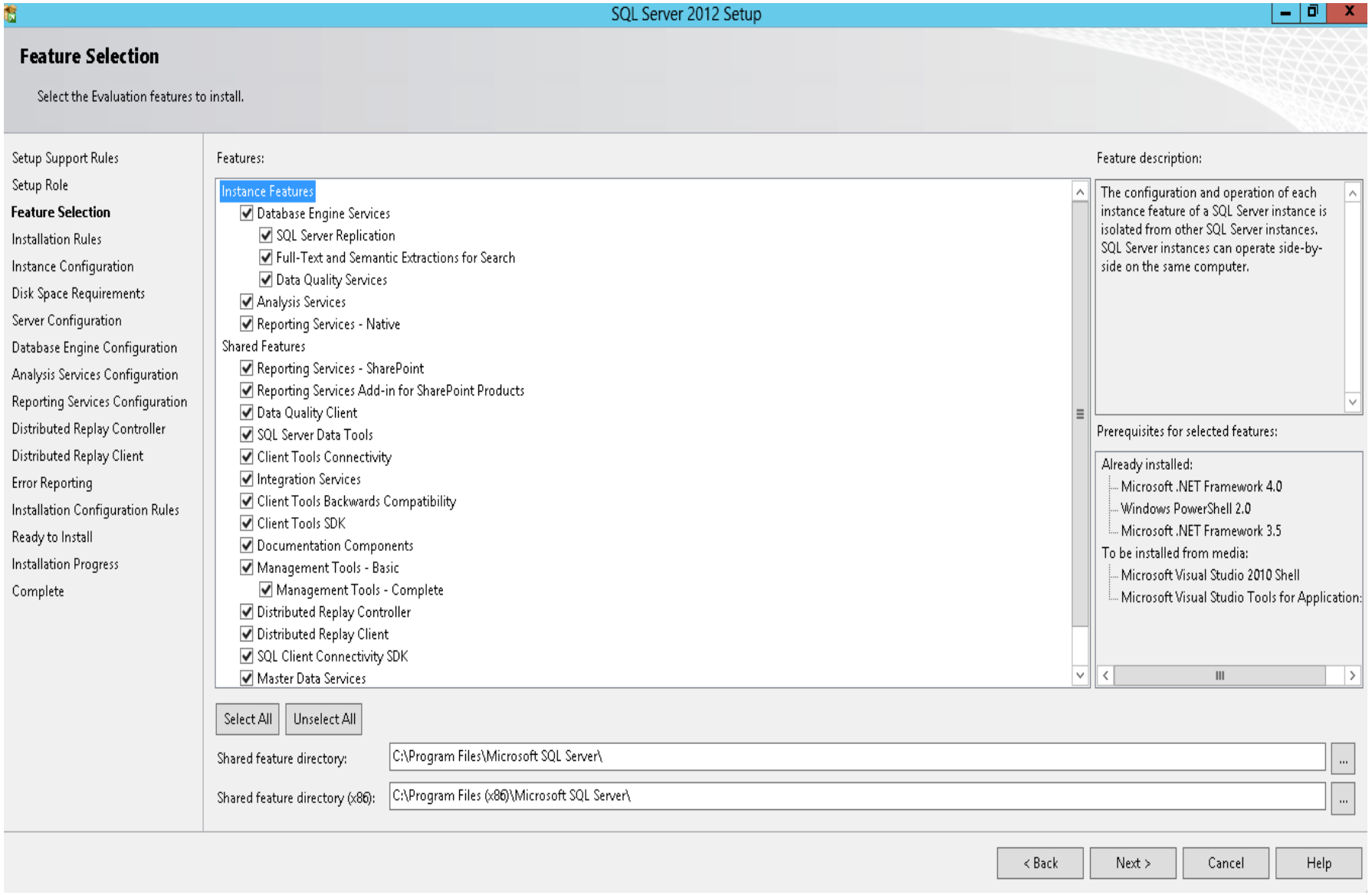
Installation – SQL Server 2012



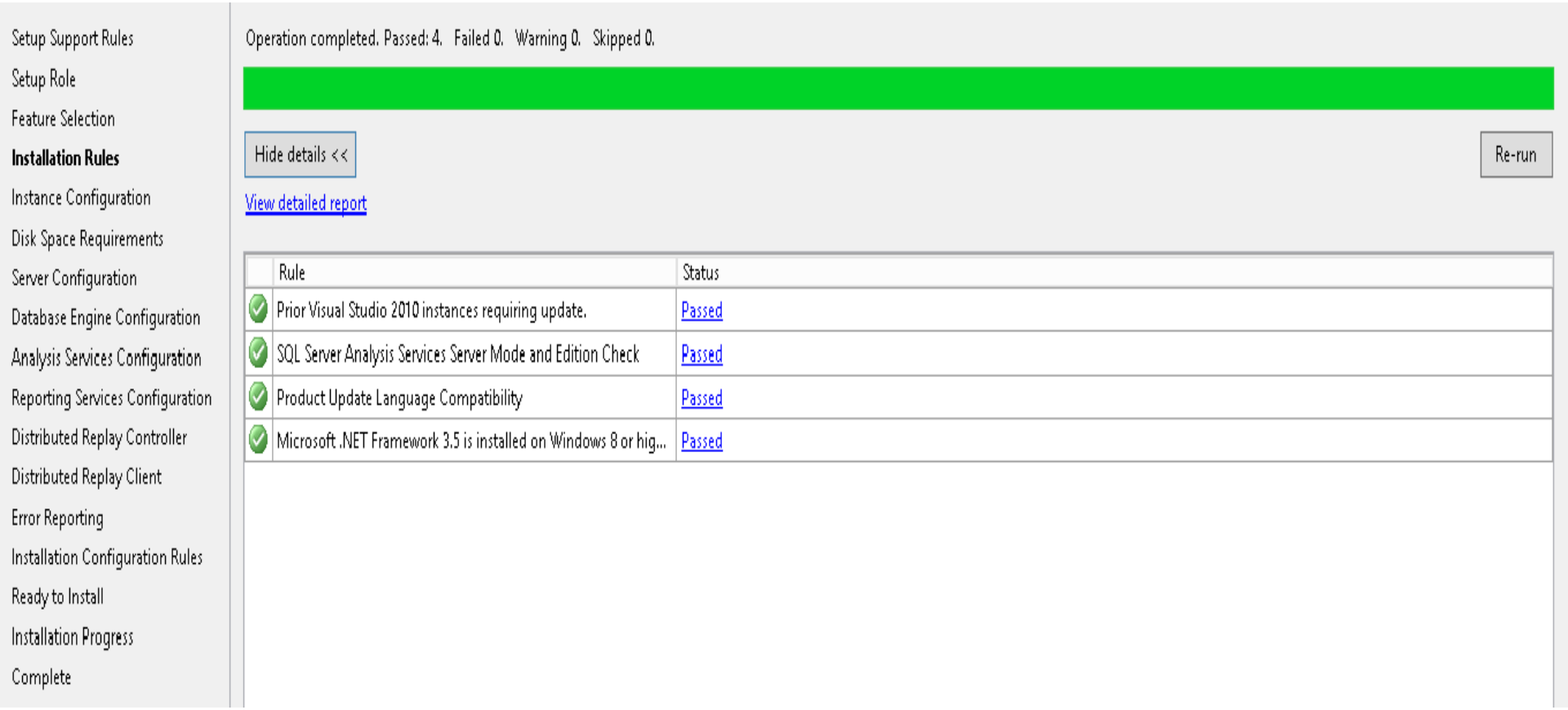
Installation – SQL Server 2012



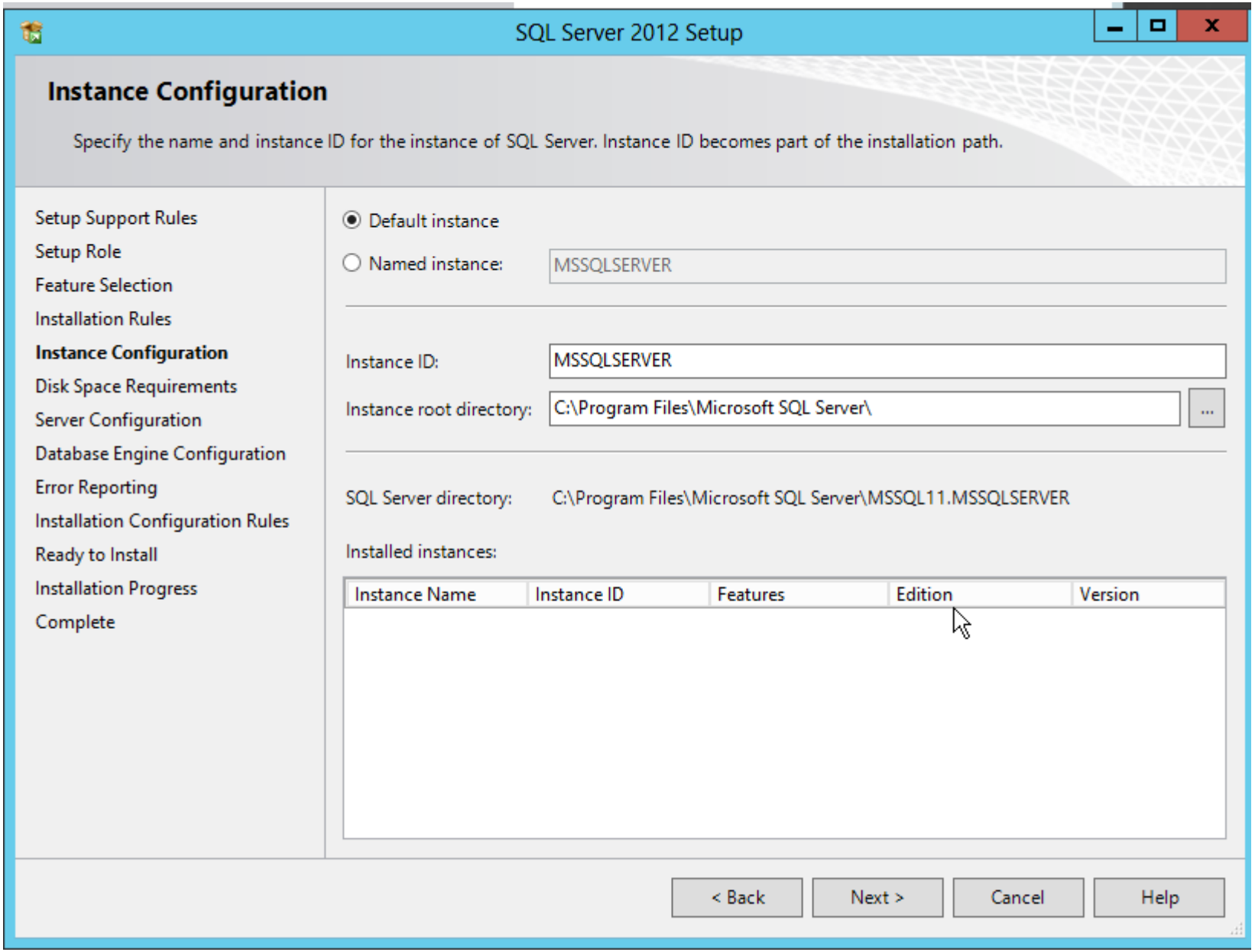
Installation – SQL Server 2012



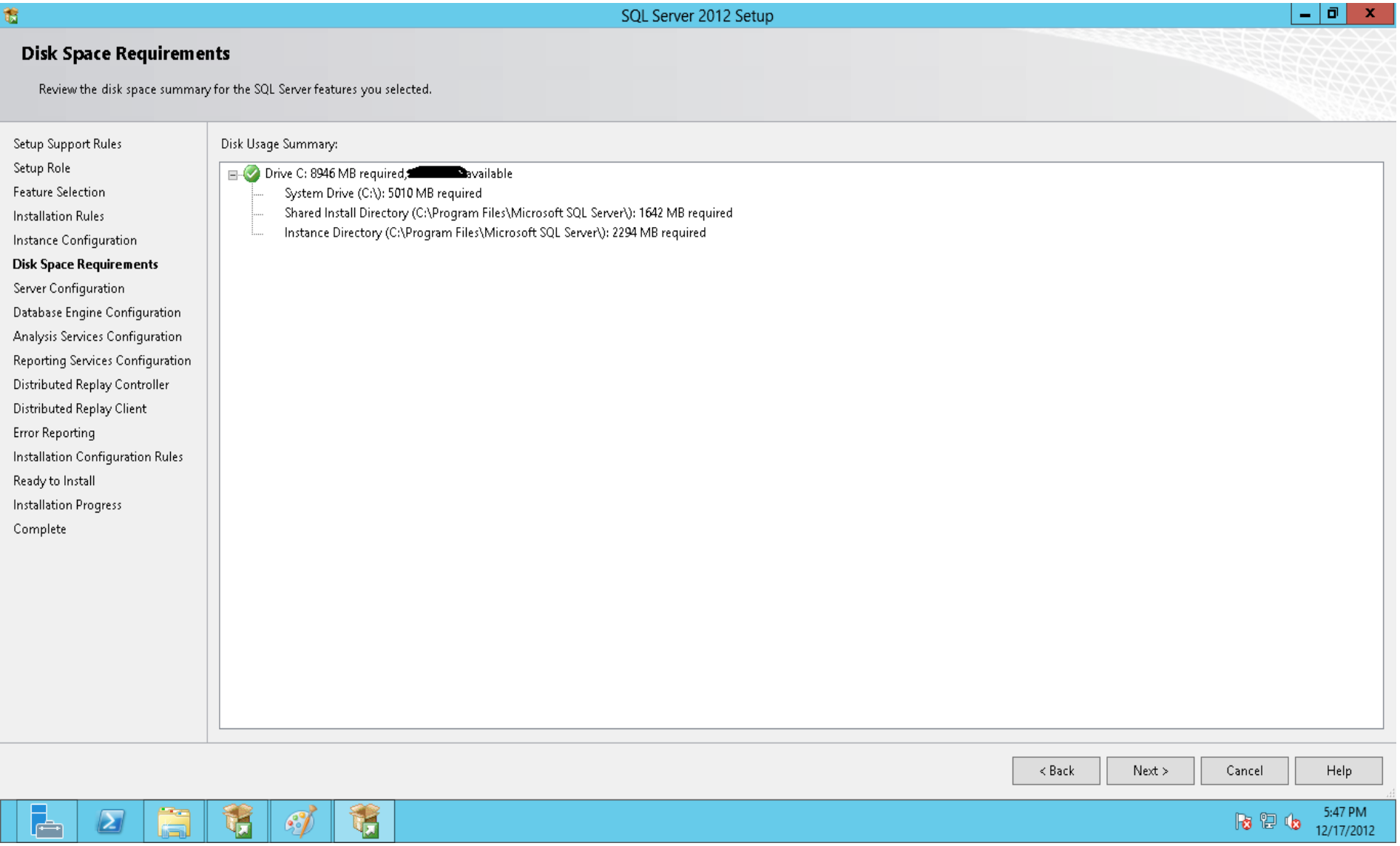
Installation – SQL Server 2012



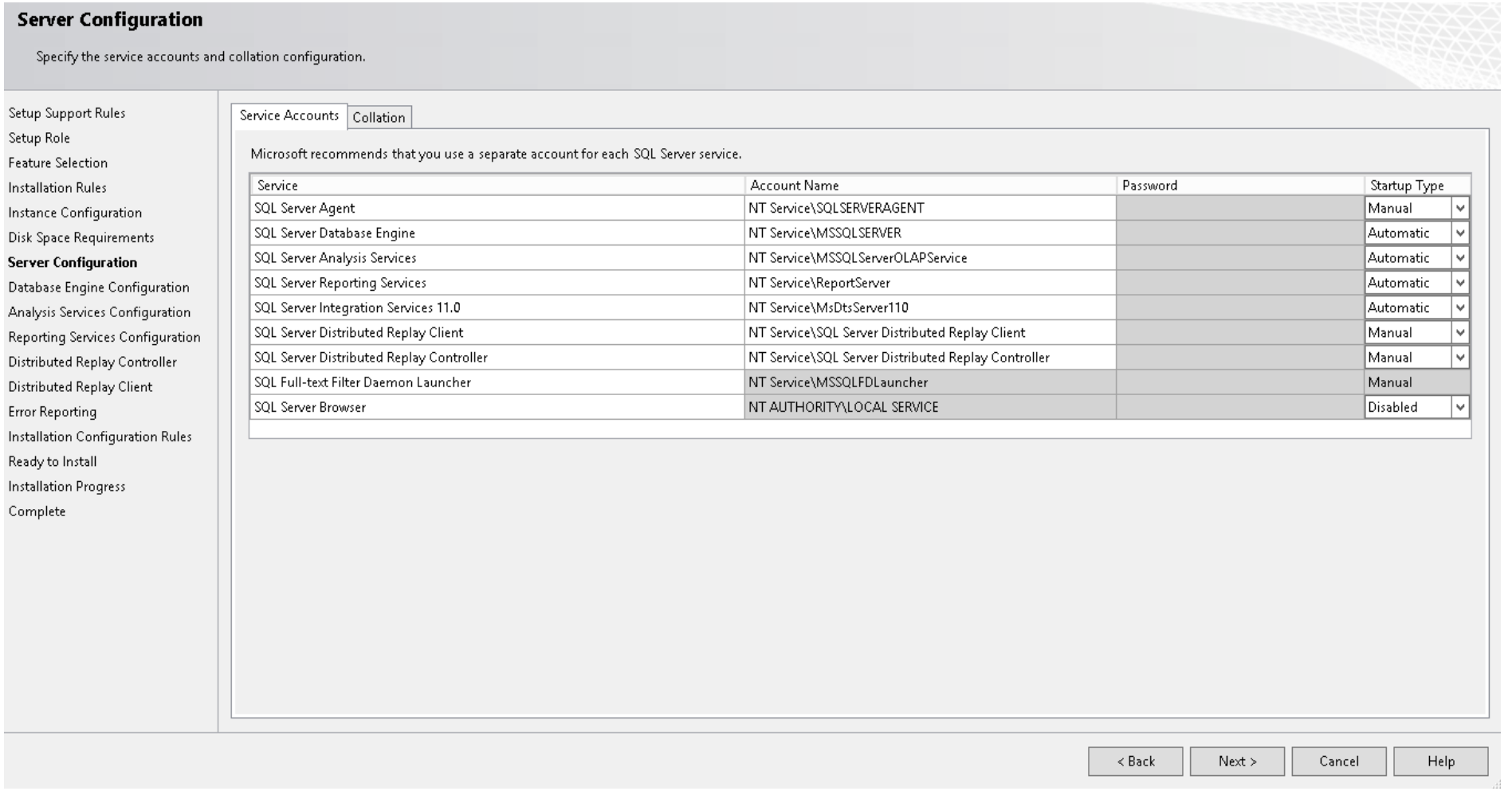
Installation – SQL Server 2012



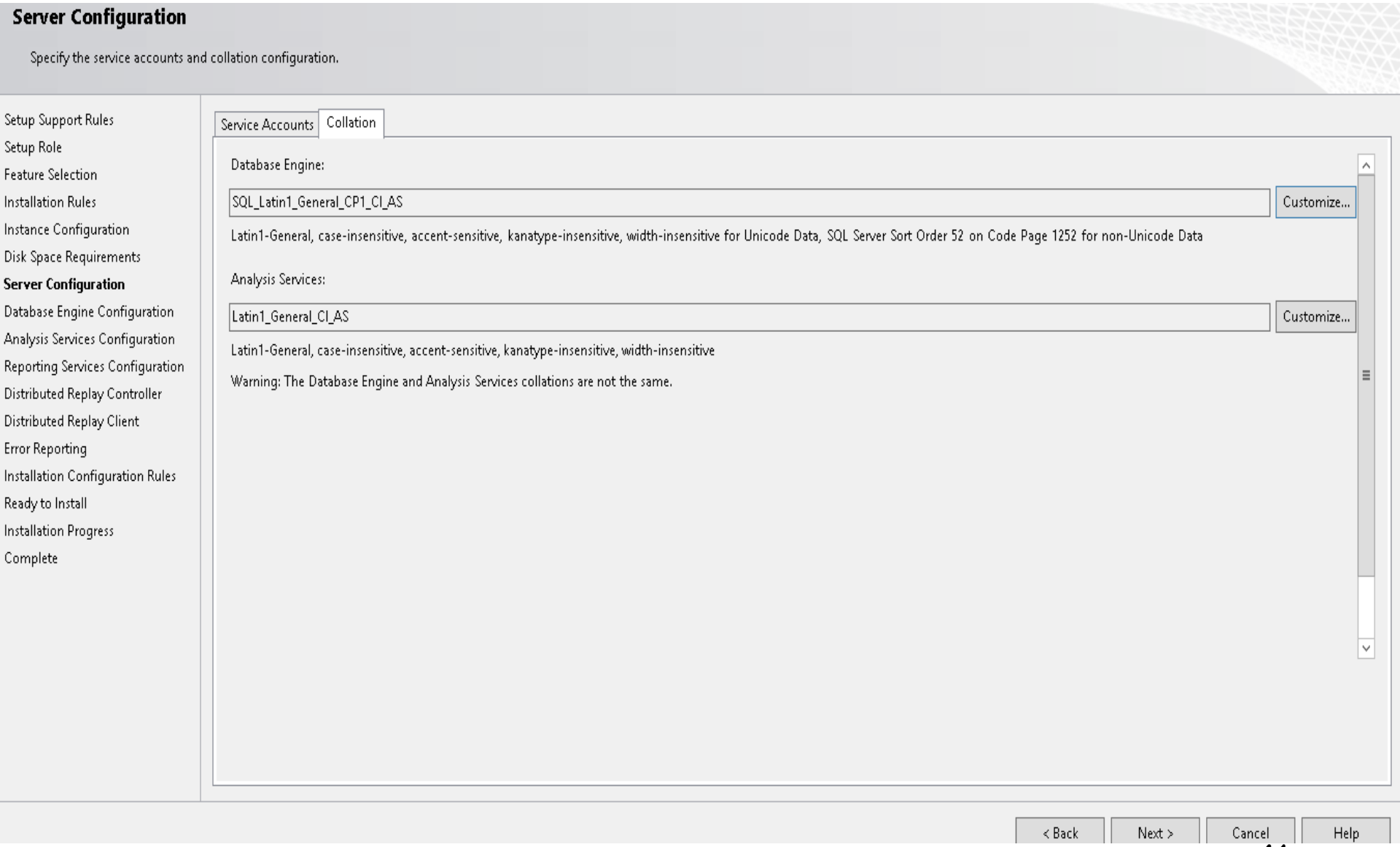
Installation – SQL Server 2012



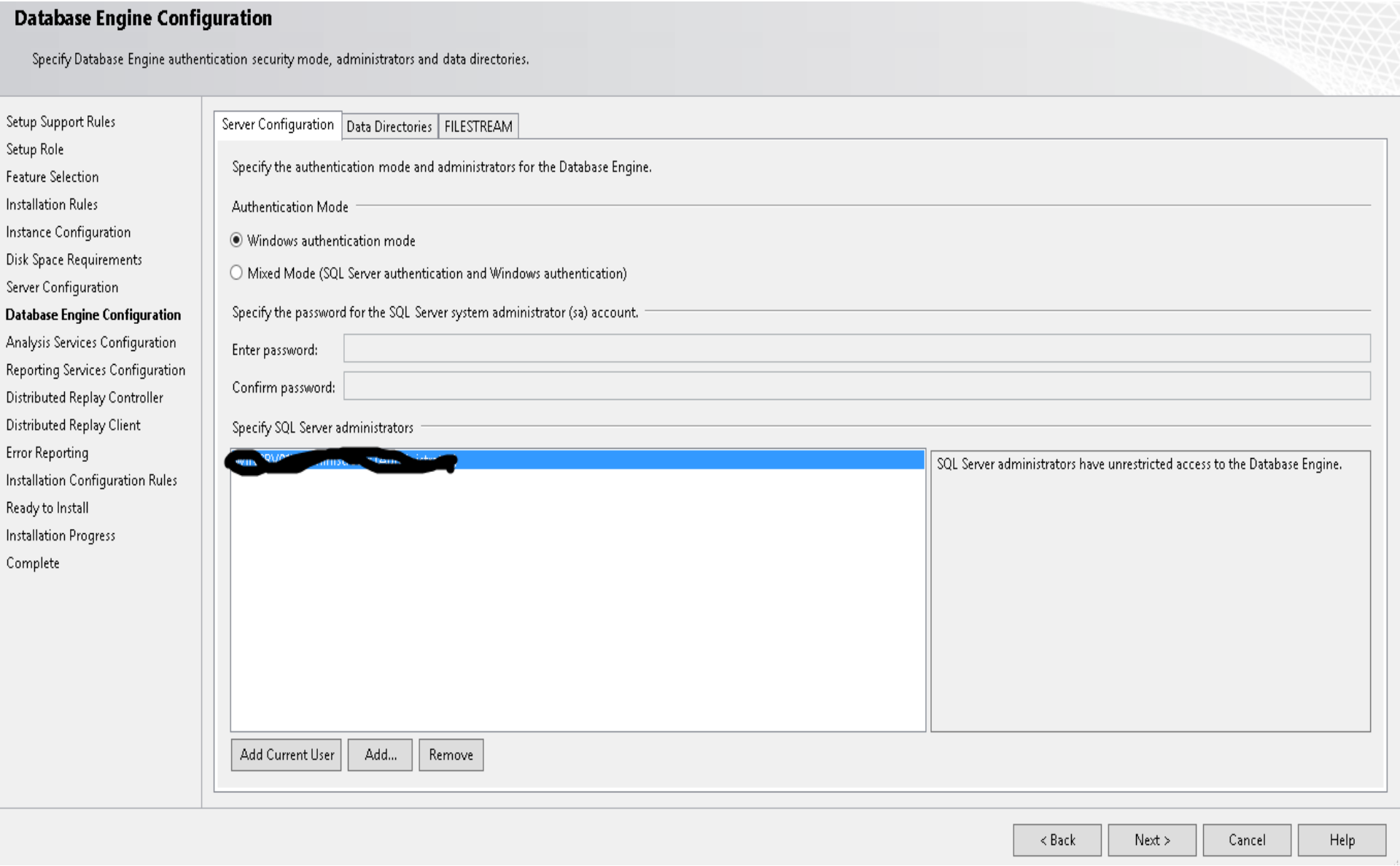
Installation – SQL Server 2012



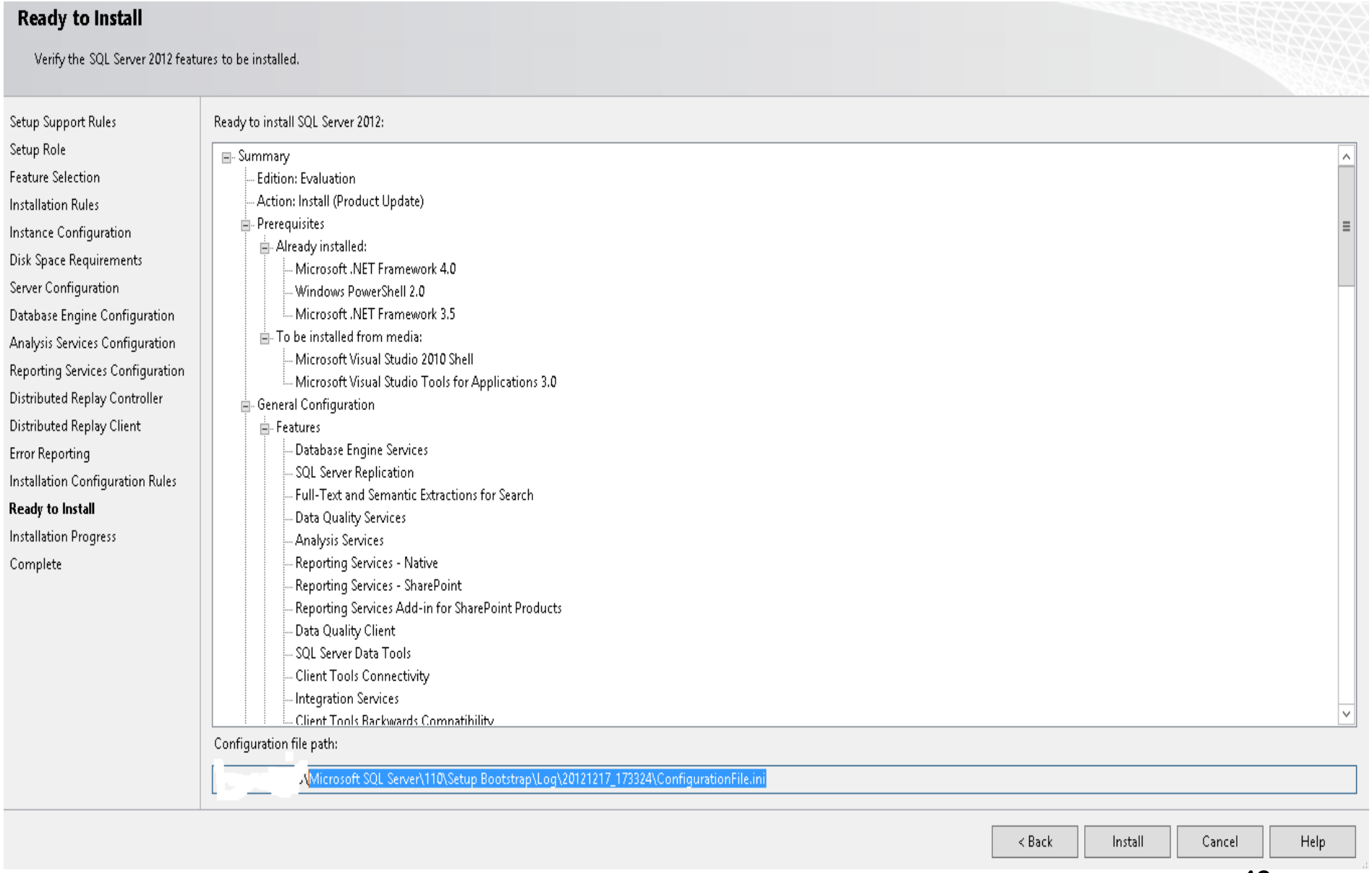
Installation – SQL Server 2012



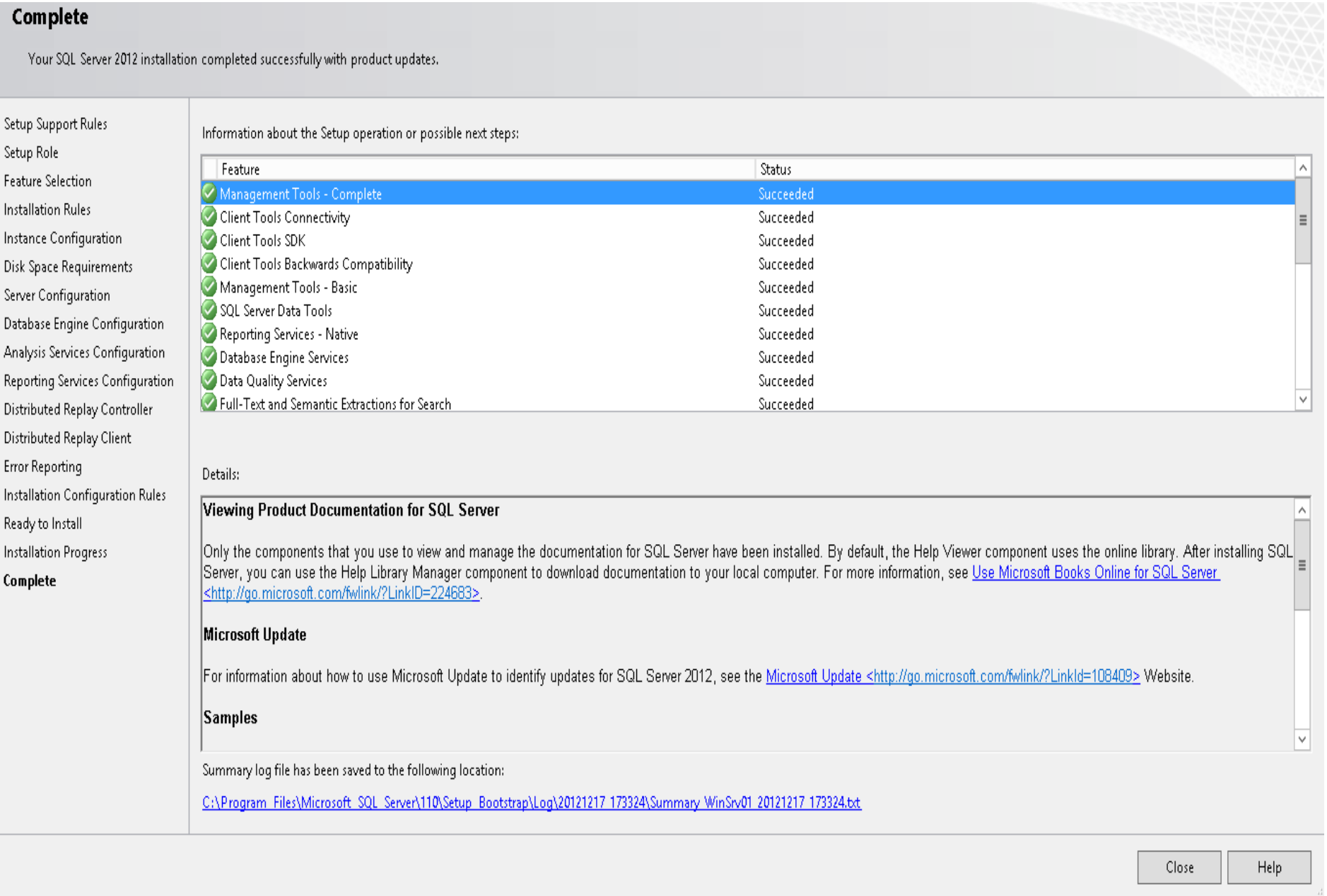
Installation – SQL Server 2012



Installation – SQL Server 2012



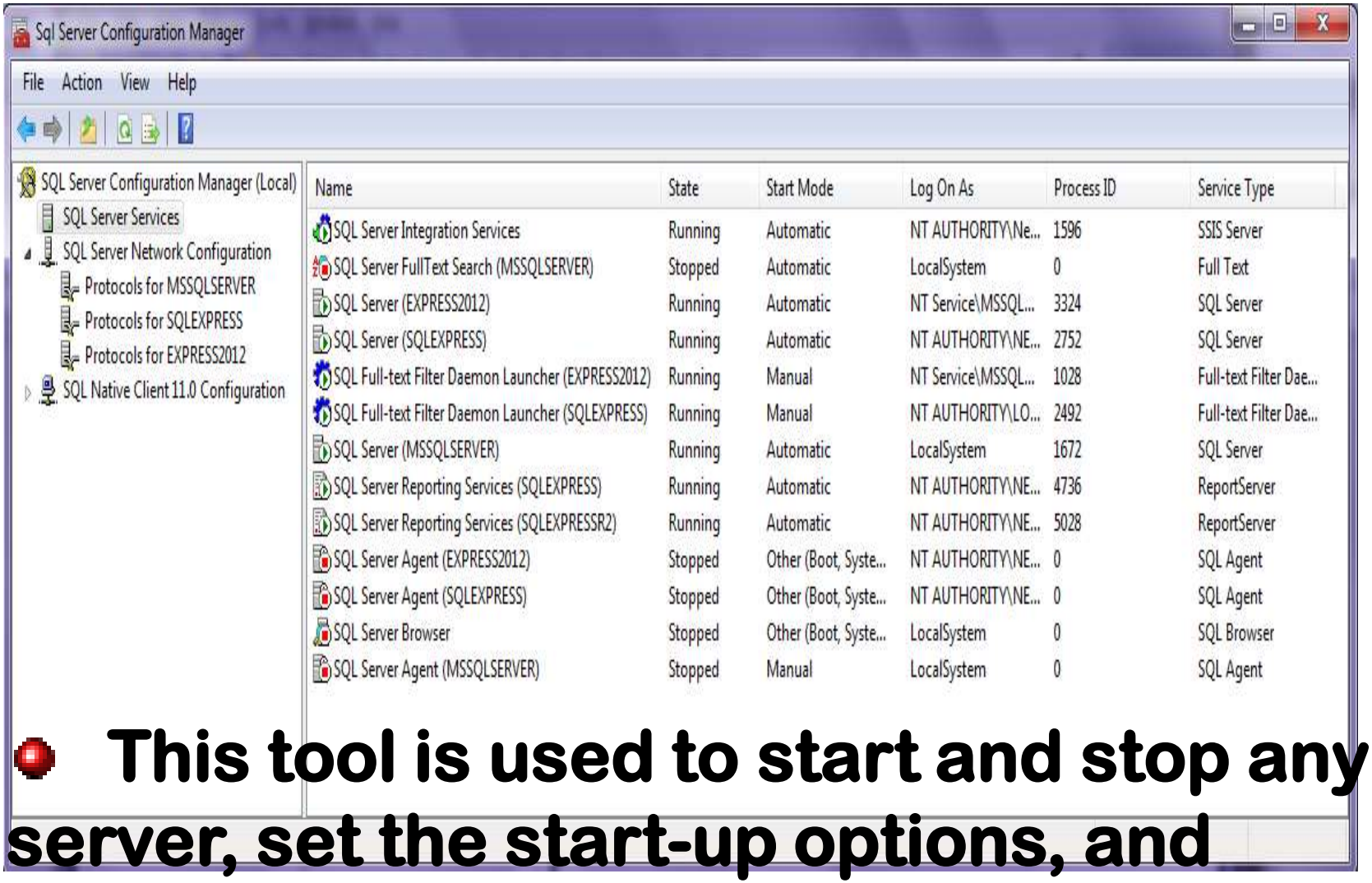
Installation – SQL Server 2012



SQL Server Tools and Components



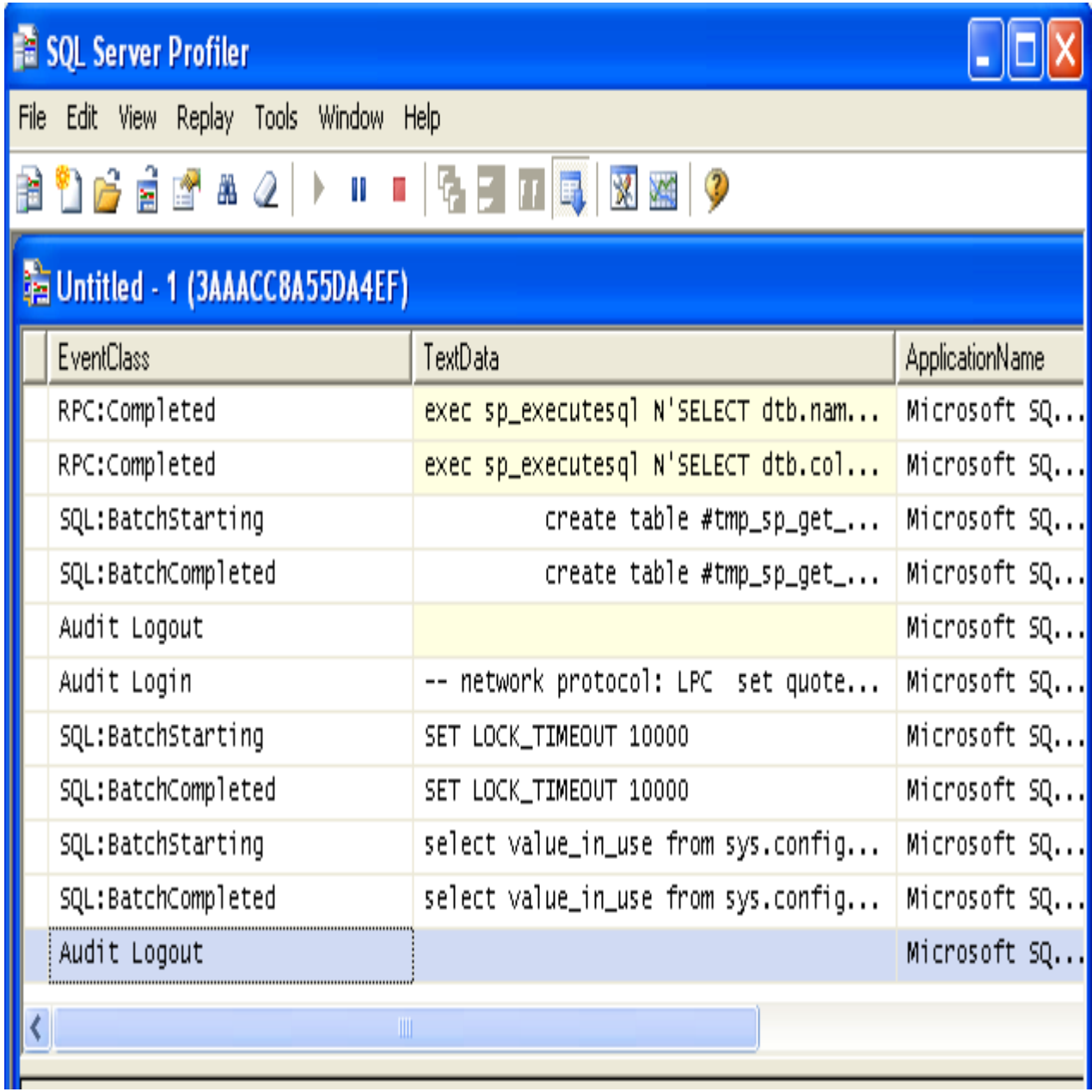
SQL Server Configuration Manager



- **This tool is used to start and stop any server, set the start-up options, and configure network protocol and manage network connectivity configuration**

SQL Profiler

- SQL Server is capable of exposing a trace of selected events and data points.
- Profiler is great for debugging an application or tuning the database



Logging In

- **Server Type**
- **Server Name**
- **Authentication**



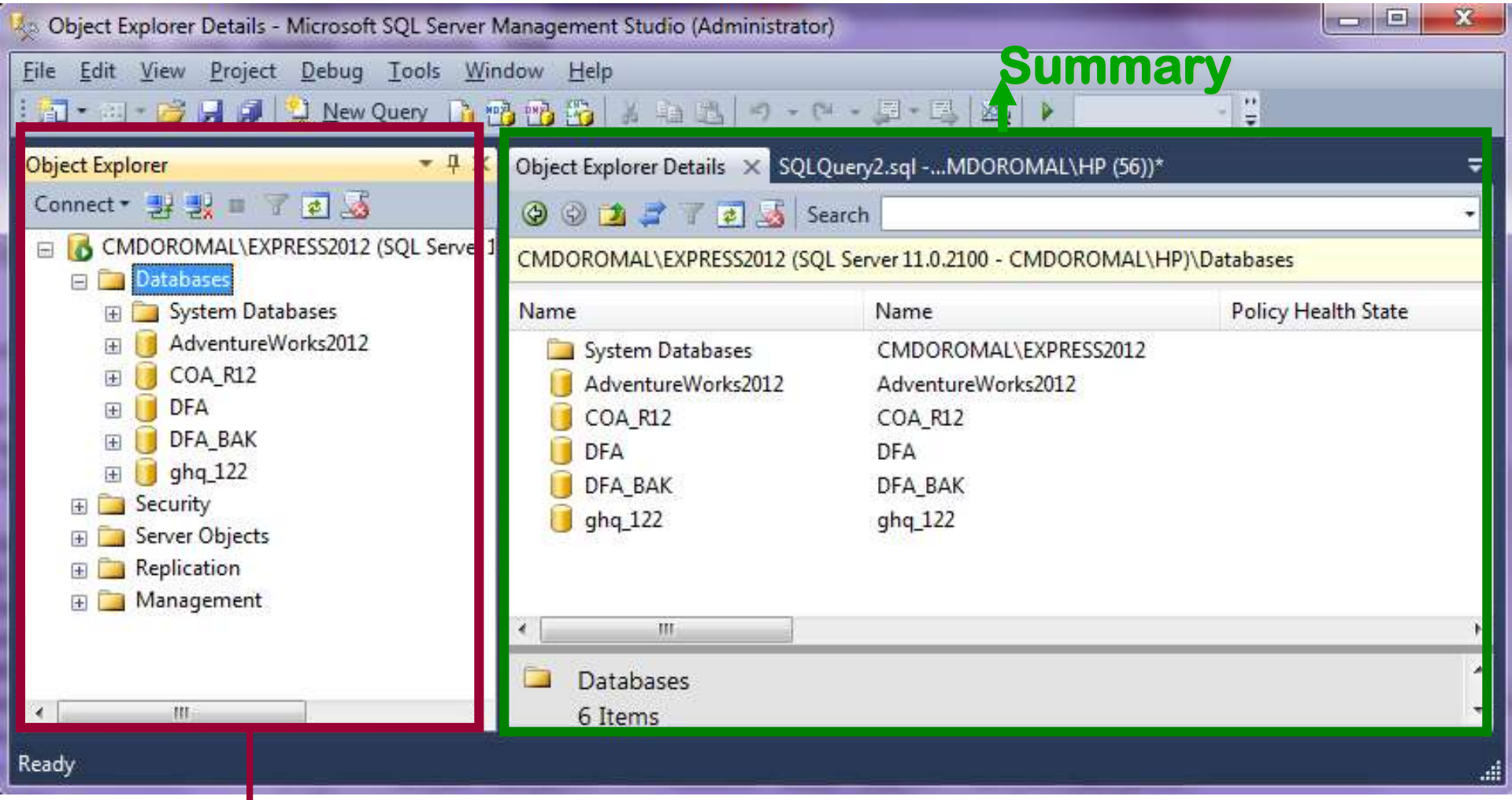
SQL Server Management Studio

- **is the primary graphical tool that performs many of the administrative tasks needed in SQL Server.**

- **enables the developer or DBA to develop database projects and manage SQL Server with either a GUI interface or T-SQL code**

- **Management Studio's Query Editor is an excellent way to work with raw T-SQL code and it's integrated with the Solution Explorer to manage projects**

SQL Server Management Studio



Object Explorer

SQL Server Management Studio

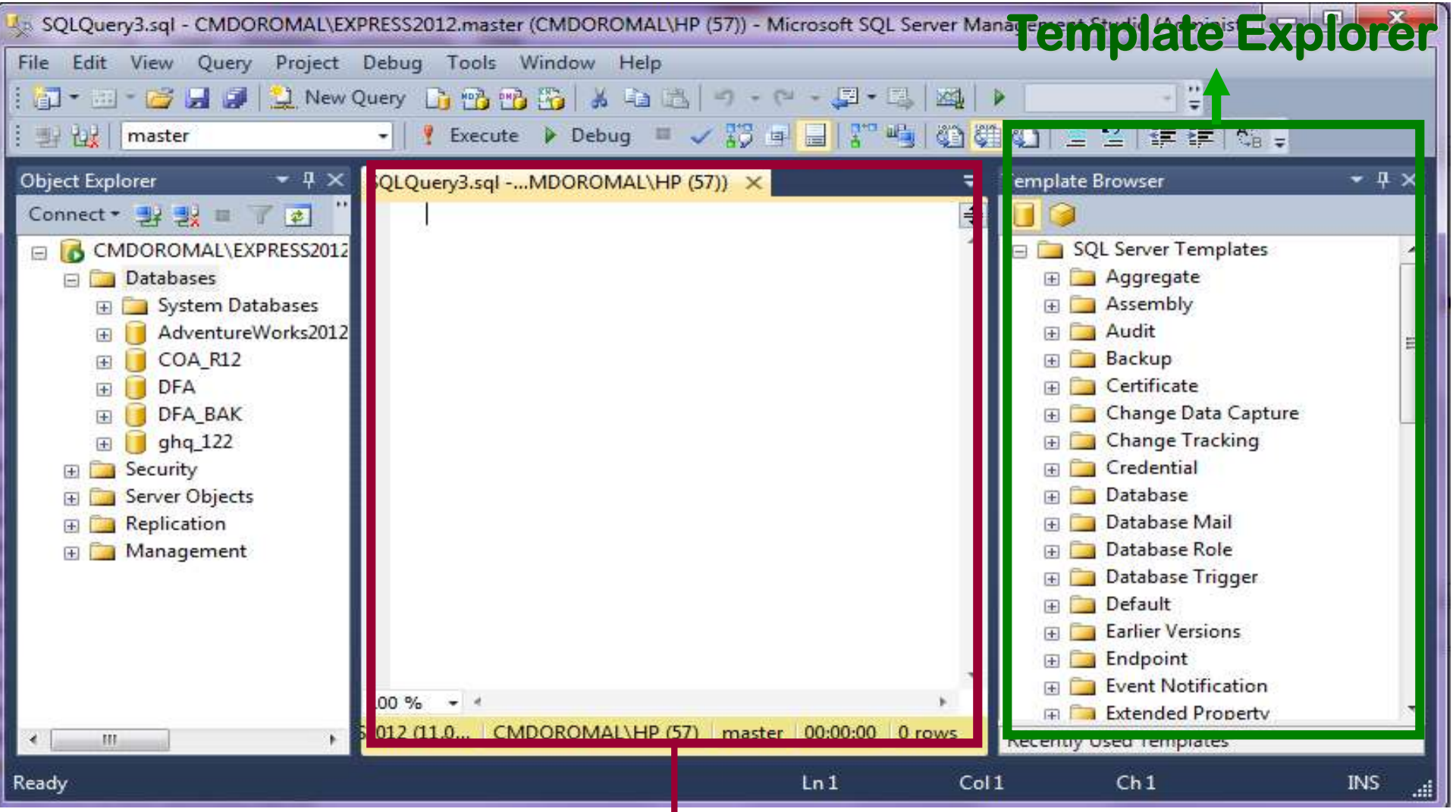
Object Explorer

- Used for administering and developing SQL Server 2008 database objects

Summary

- Presents basic information about the selected object

SQL Server Management Studio



Template Explorer

Query Editor

SQL Server Management Studio

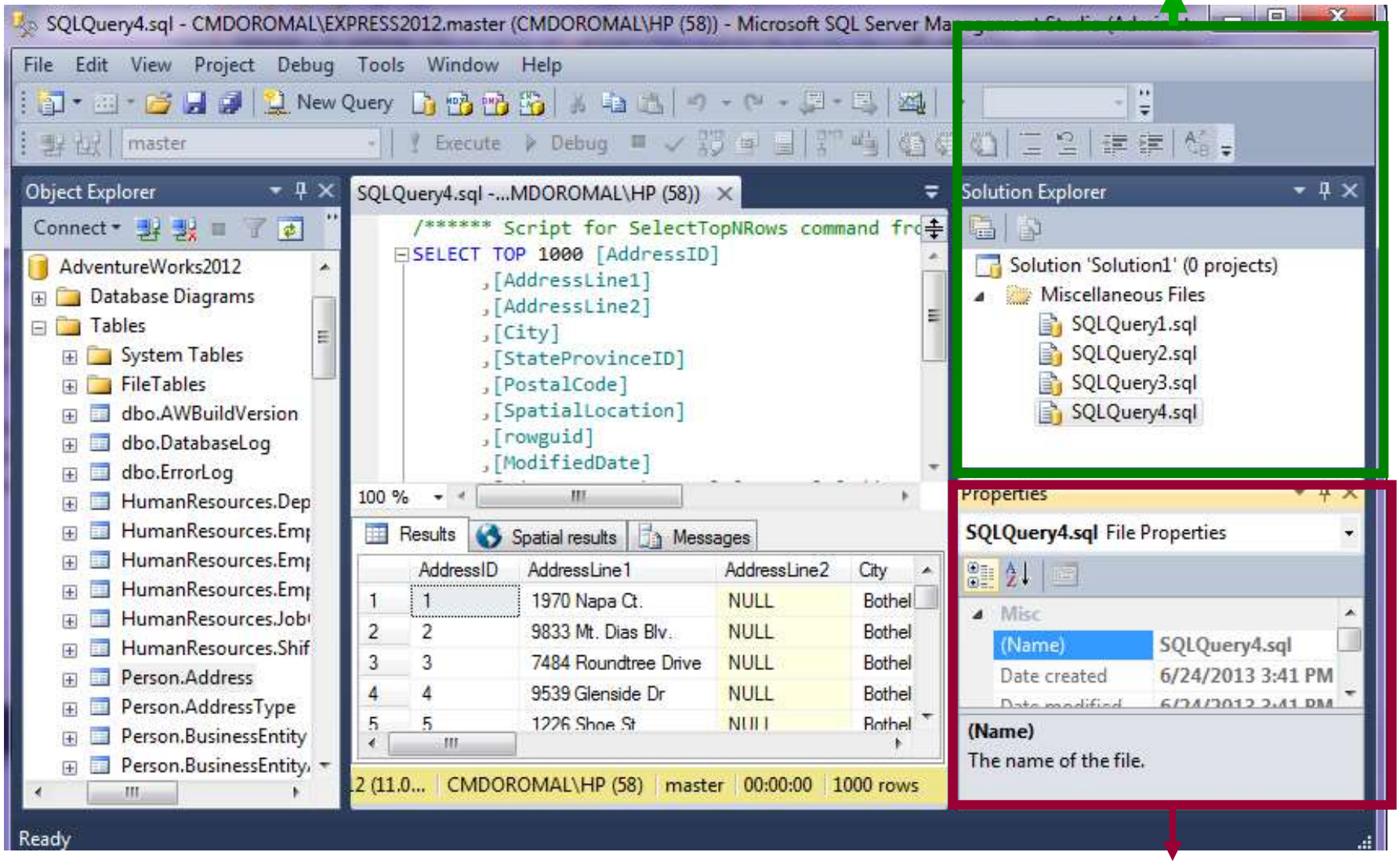
Template Explorer

☞ Used to create and manage T-SQL code templates. The templates are simply stored SQL scripts within a directory structure

Query Editor

is used to create, edit, and execute T-SQL batches.

SQL Server Management Studio



SQL Server Management Studio

Solution Explorer

Displays solutions, which are convenient groupings of objects, T-SQL, or special programs called stored procedures, among other items

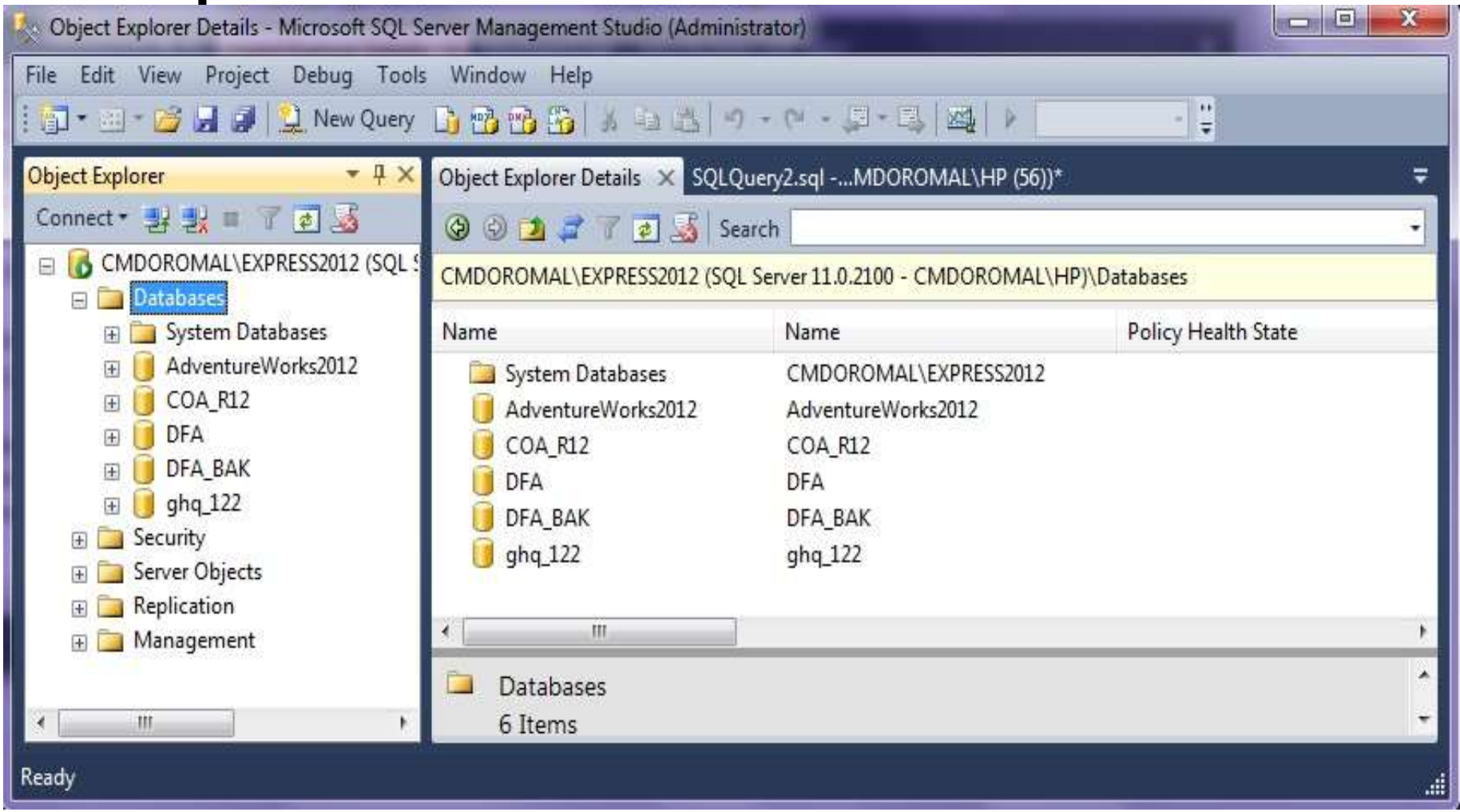
Properties Window

Displays the set of properties for each object

Viewing Available Databases

Object Explorer

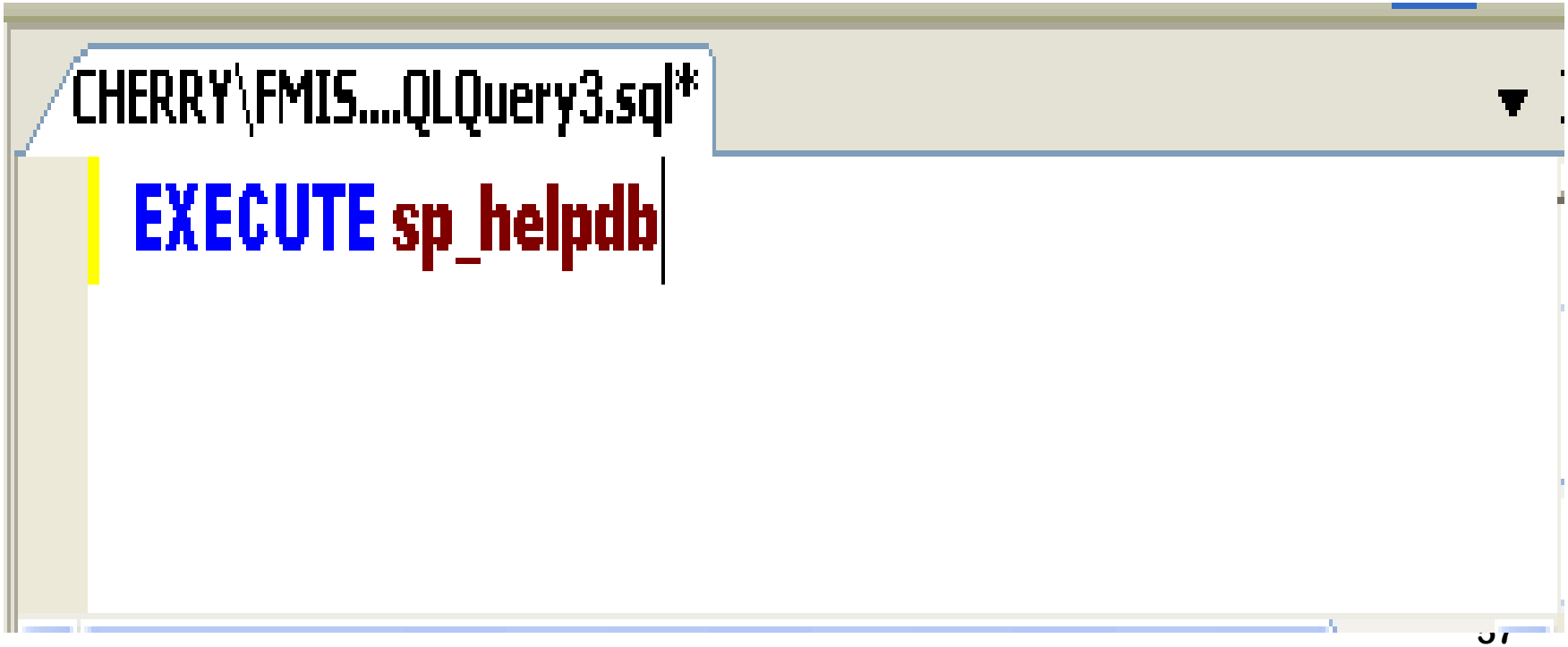
- Expand Databases folder



Viewing Available Databases

Query Editor (T-SQL)

- Execute sp_helpdb system stored procedure



System Database

- **Some databases are created automatically when you install SQL Server.**

- **The default system databases should not be deleted because they may cause failure to your instance of SQL Server.**

Master

- keeps the instance of SQL Server functioning.

- This database records all the system information for the SQL Server instance

Master

- **contains all information that is global to the server, including logins, error messages, system configuration, and the list of other databases that exist on the server**

Msdb

- **store information about the scheduling of alerts, the definition of jobs, and the recording of the server operators to be contacted when a particular event occurs on the server**

Tempdb

- contains all temporarily created stored procedures and tables and is generally used as a work area by SQL Server

- Tempdb is where tasks that require memory are performed, such as join and sort operations

Model

- **stores a complete template for creating new databases**

- **When you create a new database, SQL Server copies the whole contents of the Model database into the new database you create**

Sample Database

TSQL2012

